
System1Bench: Reference-Aware Paired Audits of Typed Decisions Across Domains

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Abstract

Aggregate accuracy can conceal changes in the individual decisions consumed by an application. We present System1Bench as a reference-aware, paired audit of typed decision interfaces, not a single cross-domain leaderboard or an end-to-end workflow test. Fifteen historical source collections provide heterogeneous decision functions and reference types; separate frozen diagnostics use executable policies and independently sourced ContractNLI and SciFact records. On Banking77, reversing the candidate dictionary changes 38% of Laya Multilingual decisions with no net accuracy change. In the legal diagnostic, the same intervention raises two code-scored models' reference agreement while leaving only 33/144 and 51/144 cases correct under both renderings. In a post-hoc, same-source census of 339 three-class SciFact cited pairs, Qwen's total reference agreement changes only 231 $\rightarrow$ 220 while no-information agreement falls 112 $\rightarrow$ 68/130; the 159-pair complement to an earlier balanced selection preserves the class-level loss. A factorial codebook control reveals distinct sensitivities to display order and answer-code identity in two fixed adapters. These findings motivate reporting reference agreement, correctness turnover and robust correctness together, with source provenance and interface scope attached.

1 Introduction

An application that routes a customer request or flags an evidence claim consumes a particular selected action. Two systems with the same accuracy may fail on different cases; a single system may preserve its accuracy while changing many actions when an equivalent candidate set is rendered differently. For a typed decision interface, reliability therefore has at least two coordinates: agreement with an appropriate reference and stability under a specified intervention. Neither can be inferred from the other.

This question is not new in isolation. Behavioral tests formalize invariance and directional expectations [Ribeiro et al., 2020]; broad evaluations emphasize multiple metrics [Liang et al., 2022]; option identifiers, order and answer-selection procedures have known effects [Zheng et al., 2023, Alzahrani et al., 2024, Wang et al., 2024]. Recent legal and computational-social-science studies also compare decision models with language models across reference quality, corrections, regressions, calibration and cost [Zhang et al., 2026, Ibrahim and Zaki, 2026]. JevAdvBench already tests typed perturbations and repeated requests [Hu et al., 2026]. The contribution here is an auditable *combination*: source-specific reference provenance, matched correctness transitions across native heads and fixed language-model adapters, orthogonal interface controls, and a separate external-domain check. We do not claim to be the first cross-domain or order-sensitivity benchmark.

The historical atlas contains 15 source collections and 28 main suites; its 8 additional controls reuse source cases. Because labels range from dataset annotations to authored policies and synthetic teachers, their percentages are never averaged into one ability score. We subsequently froze two new diagnostics before their respective inference: 288 executable-policy base states, and 264 pairs from legal contracts and scientific abstracts. A separately frozen three-class SciFact extension selected 180 cases; after inspecting it, a fresh, post-hoc census covered all 339 eligible cited pairs. None is silently counted as a historical suite. The model-facing object throughout is a supplied state and typed question map. Our unit is a *model plus adapter on a fixed downstream decision*, not a deployed legal, scientific or customer-service system.

Three empirical observations structure the paper. First, exact repeats and candidate reversals expose prediction changes that net accuracy can hide. Second, independent legal and scientific sources show that correctness turnover varies across source–task–adapter configurations; the three-class science check further exposes cancellation across reference classes. Third, a factorial codebook control demonstrates that ordinary reversal in a single-token language-model adapter jointly changes semantic display and code assignment. The final section examines prompting, source validity and statistical boundaries rather than treating any favorable rendering as a universal repair.

2 A source-stratified paired design

Typed workload and reference. Each case supplies state s_i and a question map Q_i containing an instruction, a typed output primitive and, where appropriate, named candidates or an ordinal rubric. The primitives are finite *choice*, binary probability (nou1), and ordered *score*. Only state and questions are sent to a model; labels, source IDs and other metadata remain outside the inference payload. Table 1 separates historical source families from the two independent natural-domain extensions. Function tags overlap, and source labels, authored cases, synthetic teachers and executable policies are distinct reference standards. In particular, agreement with a teacher is not independently established correctness.

Stratum	Source collections and typed functions	Reference
Historical semantic/evidence	AG News, Emotion, SST-5, BoolQ, XNLI, TurtleBench; classification, evidence, ordinal judgment	Dataset labels
Historical routing/rejection	Banking77, MASSIVE, CLINC150/OOS; intent, action or explicit rejection	Dataset labels
Historical safety/workflow	Prompt Injections, Aegis2, Typed Decisions, JevBench, ReflexBench, Jev-Laya; safety, workflow, retrieval	Dataset, human, authored, teacher or program
External legal	ContractNLI; NDA-hypothesis inference	Dataset labels
External science	SciFact; claim-cited-abstract relation	Evidence labels / absence proxy

Table 1: Fifteen historical collections (first three rows) and two separately frozen external sources (last two). Functions are not exclusive; the table classifies workloads, not system quality. Full suite-level mapping and source provenance appear in the supplement.

Paired quantities. Let y_i denote the reference and $\hat{y}_i^{(a)}$ the canonicalized prediction under condition a , counting invalid answers as incorrect. Write $c_i^{(a)} = \mathbf{1}[\hat{y}_i^{(a)} = y_i]$. For original o and intervention v , the accuracy change, prediction discordance, correctness turnover and robust-correct rate are

$$\Delta = n^{-1} \sum_i (c_i^{(v)} - c_i^{(o)}), \quad D = n^{-1} \sum_i \mathbf{1}[\hat{y}_i^{(v)} \neq \hat{y}_i^{(o)}], \quad (1)$$

$$T = (N_{01} + N_{10})/n, \quad J = n^{-1} \sum_i c_i^{(o)} c_i^{(v)}. \quad (2)$$

Here N_{01} and N_{10} count incorrect-to-correct and correct-to-incorrect transitions. Thus $\Delta = (N_{01} - N_{10})/n$ can vanish while T is large; D additionally records changes between two incorrect labels. A byte-identical repeat estimates observed execution variation, not by itself a causal order effect. We preserve case identities so all comparisons are paired.

Systems and interventions. We compare Laya English and Multilingual native typed heads, two fixed 8B instruction models (Llama 3.1 and Qwen3) using distinct single-token answer-code logits, and hosted Jev 1.13 [Convai Innovations, 2026, TypeSafe AI, 2026]. The candidate-code probabilities are conditional on that code set, not calibrated equivalents of native head probabilities. Historical order and encoding contrasts are exploratory because their outputs preceded this analysis. The

fresh executable diagnostic fixes 288 base states across three policy families, with same-request, order, language, distractor, paraphrase, encoding and one-field fact controls; a separate codebook run independently varies display order and answer-code identity. The legal/scientific diagnostic freezes 144 ContractNLI and 120 SciFact source pairs, each in base, exact-repeat and reversed-candidate conditions. Each contrast changes the declared interface, not the source reference. We use paired source/state-cluster bootstrap intervals (10,000 draws) and distinguish pointwise exploratory intervals from the within-family simultaneous policy intervals. Full seeds, exact request hashes, raw predictions and input-completeness checks are retained in a companion analysis artifact; reviewer access and upstream-license compliance still require an approved plan.

3 What historical controls establish

The historical collection is an atlas of decision functions, not a pooled test set. In four candidate-order controls, changing the presentation produces heterogeneous prediction turnover even when the reference labels and underlying state are fixed. Banking⁷⁷ is particularly diagnostic: Laya English changes 48% of selected intents under reversal while gaining nine percentage points in reference accuracy; Laya Multilingual changes 38% with *zero* net accuracy change. Llama changes 65% under reversal versus 4% under same-order repeat and loses 25 points. The exact transition counts and cluster intervals are in Appendix A. Neither unchanged accuracy nor a small error-rate difference is evidence of per-case stability. Since reversal may also reassign answer tokens for the LLM adapters, we call it a *joint interface* intervention until the orthogonal codebook test in Section 5.

Historical labels vary in authority. An annotation measures agreement with its dataset’s task convention; an authored or synthetic teacher label does not establish independent ground truth. To test a stronger reference type, we froze 288 executable-policy base states, 96 each for refund, access and routing, before running the specified variants. Jev is correct on 100% of original and jointly perturbed refund and access actions, but on only 54.2% of original and 29.2% of jointly perturbed routing actions. Thus low turnover can coexist with failure to apply the supplied rule. The fresh diagnostic also includes one-field counterfactuals, distractors, wording, language and output-encoding changes. We report the full six-effect family and its within-family simultaneous intervals in the supplement, not a selected favorable intervention as an overall benchmark score.

One post-hoc error inspection suggested that Jev confused a routing conjunction with disjunction, but a separately generated held-out wording test did not yield a universal fix: explicit wording improved 7/48 to 13/48 English exclusive-or cases while reducing the analogous Chinese cell from 46/48 to 13/48. This negative check is useful because it prevents turning an error pattern into an unsupported repair mechanism. It motivates a domain extension whose references originate outside our authored policies.

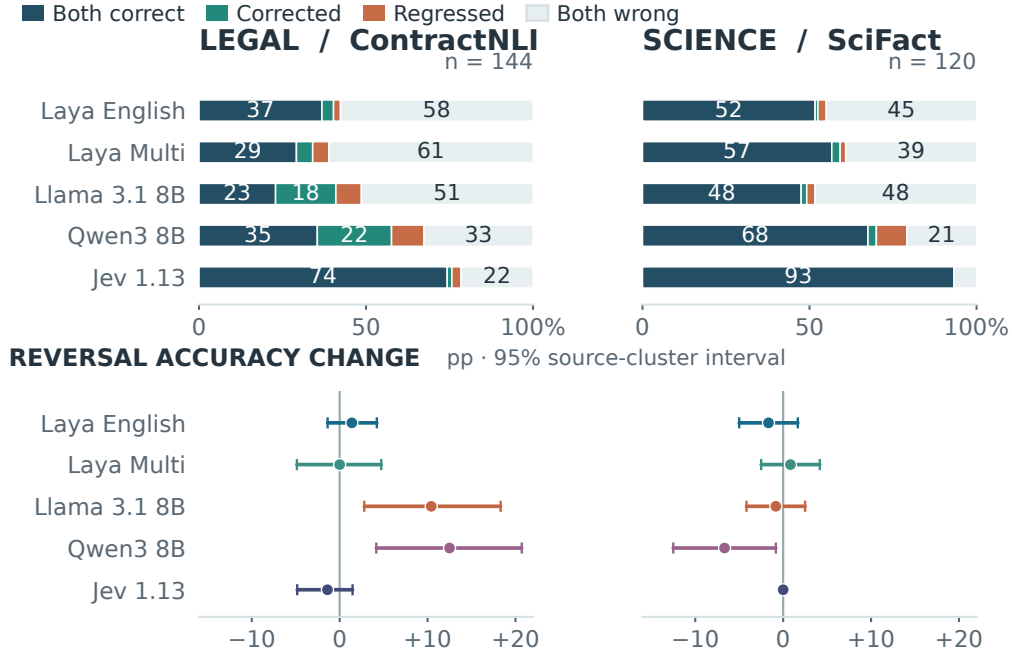
4 Independent legal and scientific references

We froze a model-blind external-domain check using 144 document–hypothesis pairs from 91 ContractNLI test contracts [Koreeda and Manning, 2021] and 120 claim–cited-abstract pairs from SciFact’s annotated development set [Wadden et al., 2020]. The legal sample has 48 cases per entailment, contradiction and not-mentioned label; the science sample has 60 support and 60 contradiction pairs. Both are intentionally balanced for paired sensitivity, not representative of natural label prevalence. Every pair receives original, byte-identical repeat and reversed-candidate requests; references and semantic candidate meanings remain unchanged. All 792 unique requests passed a complete-input audit for each of four local adapters. The hosted HTTP payloads were complete, although server-side tokenization is unobservable. Intervals below resample by contract or claim, respectively, and are pointwise unless stated otherwise.

Figure 1 partitions each domain’s cases by correctness under both renderings. In law, the code-likelihood LLM adapters change 66/144 Llama and 64/144 Qwen semantic labels. Their reference agreement rises from 44 to 59 and from 65 to 83 out of 144, respectively, but only 33 and 51 cases are correct *under both* renderings. Those net gains hide 26 corrections and 11 regressions for Llama, and 32 corrections and 14 regressions for Qwen. Pointwise contract-cluster intervals for the paired gains are [+2.8, +18.3] and [+4.2, +20.7] percentage points. These are not improvements in legal

PAIRED CORRECTNESS / LEGAL & SCIENTIFIC EVIDENCE

Same source pair · candidate reversal · five fixed model-adaptor systems



Selected balanced pairs: legal n=144 / 91 contracts · science n=120 positive cited abstracts

Figure 1: Paired correctness under ordinary candidate reversal on selected, class-balanced ContractNLI and SciFact source pairs. Each bar partitions the same pairs into both-correct, corrected, regressed and both-wrong cases; lower panels show reversal-minus-base accuracy with pointwise 95% source-cluster bootstrap intervals. The LLM intervention jointly changes display and code assignment; the science sample excludes retrieval/no-information.

reasoning: ordinary reversal changes answer-code assignment in the adapters, and Section 5 isolates that confound.

On scientific evidence, the direction need not match. Qwen loses eight correct decisions (92/120 → 84/120, -6.7 points, claim-cluster interval $[-12.5, -0.8]$), while Jev is reference-correct on the same 112/120 pairs under both orders. Jev’s legal original/reversal agreement is 138/144, compared with 142/144 on a literal repeat. These selected-pair observations are not a ranking of legal or scientific competence across native interfaces. The SciFact diagnostic contains only positive evidence relations attached to a cited abstract: it neither tests retrieval nor samples no-information decisions. A separately frozen three-class extension including no-information is kept distinct from this denominator.

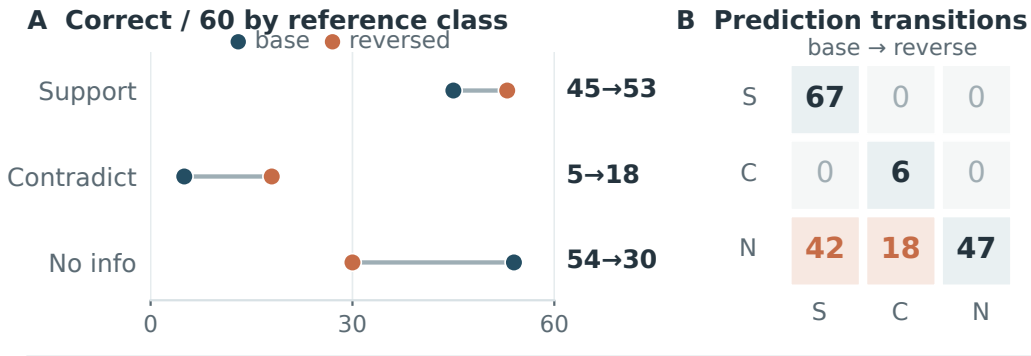
4.1 Three-way scientific evidence reveals class cancellation

The two-label SciFact check omits the no-information decision an evidence workflow may need. After inspecting that scope gap, we froze a separate, model-blind *follow-up before its own inference*, not before the earlier experiment. From 339 unique official claim-cited-abstract pairs, we selected 60 per class: 138 annotated support, 71 contradiction and 130 cited documents without annotated abstract evidence form the three eligible pools. The 180 selected claims and abstracts are each unique. NOINFO is the source’s absence-of-annotation convention, not an independently adjudicated negative. This remains a *cited-document* relation task, not open retrieval or global scientific truth. Four conditions (original, exact repeat, candidate reversal, title removal) yield 720 logical requests per system. Four local adapters passed complete-input checks.

Figure 2 sharpens the paired-accuracy argument. Qwen changes 60/180 semantic labels under reversal and none on exact repeat, yet total reference agreement barely moves, 104/180 $\rightarrow$ 101/180 (−1.7 points; paired 95% interval [−7.8, +4.4]). The net conceals a 54/60 $\rightarrow$ 30/60 drop on NOINFO (−40.0 points; label-stratified interval [−53.3, −28.3]) and a 5/60 $\rightarrow$ 18/60 rise on contradiction (+21.7 points; [+11.7, +31.7]). All 60 changed labels were base NOINFO predictions: 42 became support and 18 contradiction. This is interface-level sensitivity, not a pure semantic-order effect, because the code-likelihood adapter reassigns answer codes under reversal. Jev changes no labels under reversal, but two on exact repeat; its 148/180 base-correct cases remain jointly correct. Class-specific turnover is more informative than the small total difference, even in an intentionally balanced sample.

QWEN3 8B / 180 CITED CLAIM-ABSTRACT PAIRS

Near-flat accuracy, 60 changed semantic answers



Total correct 104 $\rightarrow$ 101/180; Δ −1.7 pp (paired 95% CI −7.8 to +4.4)

60/180 reversal flips vs 0/180 exact-repeat flips. S/C/N: support/contradict/no-info.

Figure 2: Three-class SciFact cited-abstract control for Qwen3 8B. Left: same-pair reference-correct counts by class (60/class); right: all 180 base-to-reversed semantic prediction transitions. S/C/N denote support/contradiction/no-information. Overall accuracy barely moves despite 60 label switches; exact repeat has zero. No-information is an explicitly cited abstract without annotated evidence, not an arbitrary unlabeled corpus pair. Intervals in text use 10,000 paired, label-stratified claim/document-cluster resamples.

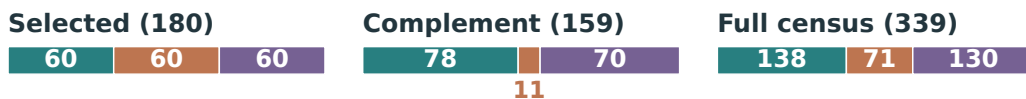
Removing the source-paper title is a different stressor: Laya English changes 60/180 labels and its reference agreement rises from 81/180 to 100/180, including 20/60 $\rightarrow$ 39/60 on NOINFO. The frozen variant removes `state.paper_title` while retaining an instruction that mentions the title, creating a minor instruction–state mismatch; a title may also contain evidence. This is neither a semantics-preserving invariance test nor proof of a shortcut. The two-class and three-class SciFact samples have different label spaces and only partial positive-case overlap, so their raw accuracies are not directly comparable. The source/reference and class-stratified audit is retained separately in the companion artifact.

Post-hoc full-census sensitivity. We reran all five systems on all 339 eligible cited claim–abstract pairs, rather than combining old predictions with new ones. In the 159 pairs outside the earlier balanced three-choice selection, Qwen’s reversal reduces NOINFO agreement 60 $\rightarrow$ 38/70 (−31.4 points; paired claim-cluster 95% interval [−42.4, −21.1]); gains on support (65 $\rightarrow$ 74/78) and contradiction (4 $\rightarrow$ 7/11) partly conceal this as 129 $\rightarrow$ 119/159 overall (interval [−13.7, +1.2]). On all 339, NOINFO is 112 $\rightarrow$ 68/130 while overall agreement is 231 $\rightarrow$ 220/339. Figure 3 shows the changed reference mix. This same-source, post-hoc check is not an independent holdout: 37 complement positives appeared in the earlier two-choice task. A separately frozen execution-context control adds 6,780 decisions: at most 7/339 predictions move per condition/plan, while Qwen’s option-reversal flips remain 95–97 and its NOINFO decline persists (Appendix D).

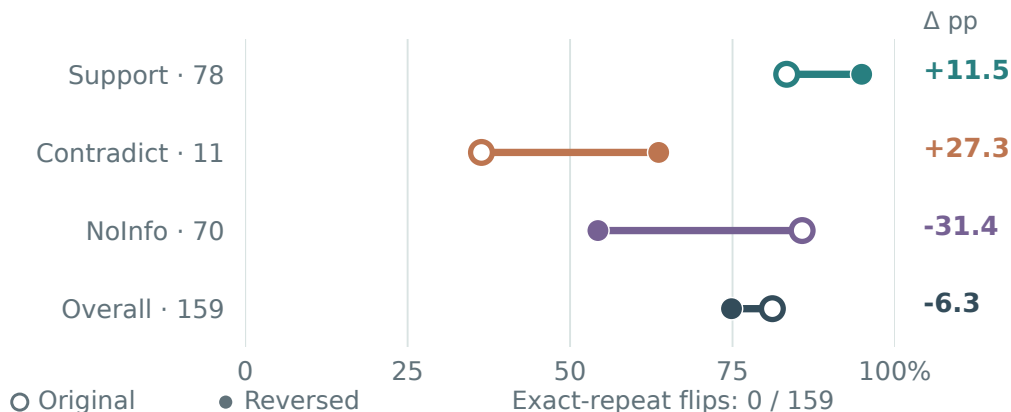
NoInfo loss survives the balanced-sample selection

SciFact claim-cited-abstract pairs · Qwen3 8B · joint option reversal

REFERENCE MIX



QWEN / 159-PAIR THREE-CHOICE COMPLEMENT



NoInfo = cited abstract without annotated evidence; not adjudicated neutrality.

Figure 3: Qwen three-class cited-pair census. The top strip contrasts the deliberately balanced 180 with its 159-pair same-source complement and the full 339. Open/filled points are original/joint-reversal reference agreement from the *fresh full run*; the printed differences are percentage points. NoInfo denotes an explicitly cited abstract without annotated evidence, not adjudicated neutrality. Reversal also remaps answer codes in this adapter; exact repeat yields zero Qwen flips on the complement. Thirty-seven complement positives appeared in the earlier two-choice task.

5 A codebook explains why reversal is composite

On the *same* 144 ContractNLI cases, we completed a 2×2 follow-up crossing semantic display order with label-to-answer-code assignment for Llama 3.1 8B and Qwen3 8B. The original and joint-reversal outputs were reused by exact request hash; the two missing single-factor cells were added with fixed model weights and adapter. Their messages and code maps were frozen before inference. A corrective amendment aligned the new batch size to the historical four after a batch-eight pilot had been inspected, and both pilots are retained. This is therefore a transparent follow-up diagnostic, not a wholly prospective factorial test.

The native accuracy grid (base, display-only, code-only, both) is 44, 48, 50, 59 correct of 144 for Llama and 65, 78, 63, 83 for Qwen; all 1,152 cell decisions are valid. But the semantic prediction transitions are almost orthogonal across these two model-adapter pairs. Code-only remapping changes 125/144 Llama labels (86.8%; 95% contract-cluster interval [81.1, 92.1]) and only 10/144 Qwen labels (6.9%; [2.8, 11.6]). Display-only reversal changes 6/144 Llama labels (4.2%; [1.4, 7.6]) and 36/144 Qwen labels (25.0%; [18.1, 32.2]). Llama keeps selecting token A in 138/144 base and 131/144 code-only cases despite remapping, whereas Qwen mostly preserves semantic answers while its preferred code token changes. This is adapter-specific coupling, not a model-family mechanism.

The historical joint reversal also changes the decoder’s priority for exactly tied BF16 logits. Figure 4 uses one *common semantic tie-break* for a comparable four-cell view, explicitly post hoc. Native counts above remain the primary reproducible record. Under this sensitivity the joint accuracy effects are +9.0 points for Llama and +13.2 for Qwen; the difference-in-differences intervals include zero in both models, so we do not claim an identified interaction. Exact repeats produce no flips. The batch-eight pilot changes three Qwen display-only predictions, underscoring the need to pin execution shape. Full tie counts and pilot details appear in Appendix H.

Order and answer-code mapping have different effects

144 matched ContractNLI pairs/model · fixed semantic tie-break · batch 4

Llama 3.1 8B

Qwen3 8B

	BASE MAP	SWAP MAP		BASE MAP	SWAP MAP
Orig.	30.6% BASE	34.7% 125 flips	Orig.	45.1% BASE	43.8% 10 flips
Rev.	33.3% 6 flips	39.6% 54 flips	Rev.	54.2% 36 flips	58.3% 69 flips

Cells show accuracy; flips count changed semantic labels vs each model's base.

Figure 4: Orthogonal ContractNLI display-order $\times$ label-to-code control on the same 144 cases per pinned 8B adapter (historical batch size four). Cells show accuracy; switches are semantic changes from base. The grid uses a fixed semantic tie-break as an explicit post-hoc sensitivity; native joint-reversal accuracies remain 59/144 for Llama and 83/144 for Qwen. This adapter-level probe does not establish a model-family mechanism.

194 A further, explicitly exploratory four-cell follow-up on the same 180 three-class SciFact cited pairs
195 shows why the factors cannot be inferred from the joint result; we focus here on Qwen, with the
196 second adapter reported in the supplement. The two new single-factor cells were frozen before
197 inference; historical base/joint tokenized prompts were hash-matched, all four cells used batch size
198 four, and Qwen's native and common-tie decisions agree. Its base, display-only, code-only and
199 both conditions have 104, 114, 114, 101 reference-correct decisions. Each isolated change gains
200 10/180 correct cases (+5.6 points), whereas their joint change loses 3/180 (−1.7 points). Semantic
201 label flips from base are 30, 18 and 60, respectively; the paired accuracy difference-in-differences
202 is −12.8 points (claim-cluster 95% interval [−19.4, −6.1]). This is non-additivity in one pinned
203 model–adapter–task combination, not an architectural mechanism or a universal codebook effect. The
204 follow-up was designed after the three-class result had been inspected, so its interval is descriptive
205 rather than confirmatory.

206 6 Boundaries and conclusion

207 The same benchmark score can move for reasons outside the intended decision function. On a frozen
208 212-state, 15-source challenge, brief zero-shot deliberation changes 55 Llama and 46 Qwen decisions
209 relative to one-pass code logits. Llama gains 12 net correct cases (25 corrections, 13 regressions;
210 three invalid generations), but its source-equal effect is +4.7 points with interval [−1.4, +10.6].
211 Qwen loses five net correct cases (12 corrections, 17 regressions; ten invalid). A post-hoc parser
212 would recover eight of those Qwen invalid answers; strict output compliance is the registered result.
213 This is evidence that output grammar and prompt are part of the evaluated adapter, not that one short
214 deliberation prompt bounds the model's ability (Appendix F).

215 An independent natural-domain control isolates a simpler adapter concern. On the identical 144 legal
216 and 180 three-class science pairs, greedy direct generation with the *unchanged* historical code prompt
217 and strict whole-response one-code parsing reproduces every constrained next-token code-logit label
218 for both 8B models: 648/648 valid outputs and zero paired semantic flips. This rules out a direct-code
219 parsing discrepancy on these samples, not a best-prompt advantage or probability equivalence.

220 **Source-validity audit.** ContractNLI repeats 17 fixed hypotheses; 43/48 sampled contradictions use
221 just four hypothesis IDs. A document-blind lookup trained on the official training contracts' modal
222 label per hypothesis reaches 82/144 (56.9%) on our balanced sample. This exposes a label-prior
223 shortcut, not a fair zero-shot baseline or evidence that our systems use it. The 16,000-character
224 cap excludes 24/123 legal test contracts, including 13/23 SEC-HTML contracts, so the selected

format mix is not neutral. SciFact’s two-label diagnostic excludes no-information and retrieval. In the three-way extension, NOINFO means an official cited pair without annotated abstract evidence; absence of annotation is a weaker reference than an explicit evidence relation. The reported intervals are conditional on selected or eligible cited pairs from this source and pinned adapters. Even the full cited-pair census is not a new population sample; it does not establish a prevalence-weighted legal or general scientific score, a deployed-system outcome, or causality at the neural-mechanism level. Native-head and LLM code probabilities have different semantics and should not be pooled as calibrated confidence.

The actionable contribution is consequently a measurement protocol: attach reference provenance to each source, preserve paired case identity, separate same-request variation from specified interface changes, report prediction discordance and both-correct coverage alongside accuracy, and isolate code assignment when using short-token adapters. The independent domains and negative controls show why these coordinates are needed; they do not yield a universal robustness winner. Reproducible request contracts, source/licensing notes and remaining validity audits are described in Appendix J and the companion artifact.

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290 A Historical candidate controls and source detail

291 The historical collection contains 28 main suites from 15 source collections, plus eight matched
 292 controls that reuse cases. A control is not an independent benchmark sample. Source-family labels,
 293 functions and reference types in Table 1 are expanded in the suite-level artifact. The four order
 294 controls shown below use distinct denominators and never contribute to one pooled effect.

Source	Model	Repeat flips	Reversal flips	Δ accuracy
banking77	Laya EN	0.0	48.0	+9.0
banking77	Laya Multi	0.0	38.0	+0.0
banking77	Llama	4.0	65.0	-25.0
banking77	Qwen	1.0	39.0	-18.0
banking77	Jev	1.0	3.0	+1.0
massive_en	Laya EN	0.0	50.0	-2.0
massive_en	Laya Multi	0.0	46.0	-7.0
massive_en	Llama	5.0	51.0	-2.0
massive_en	Qwen	0.0	32.0	-8.0
massive_en	Jev	2.0	10.0	-1.0

Table 2: Matched candidate controls. Flip rates and accuracy differences are percentage points; reversal is compared with the corresponding same-order repeat.

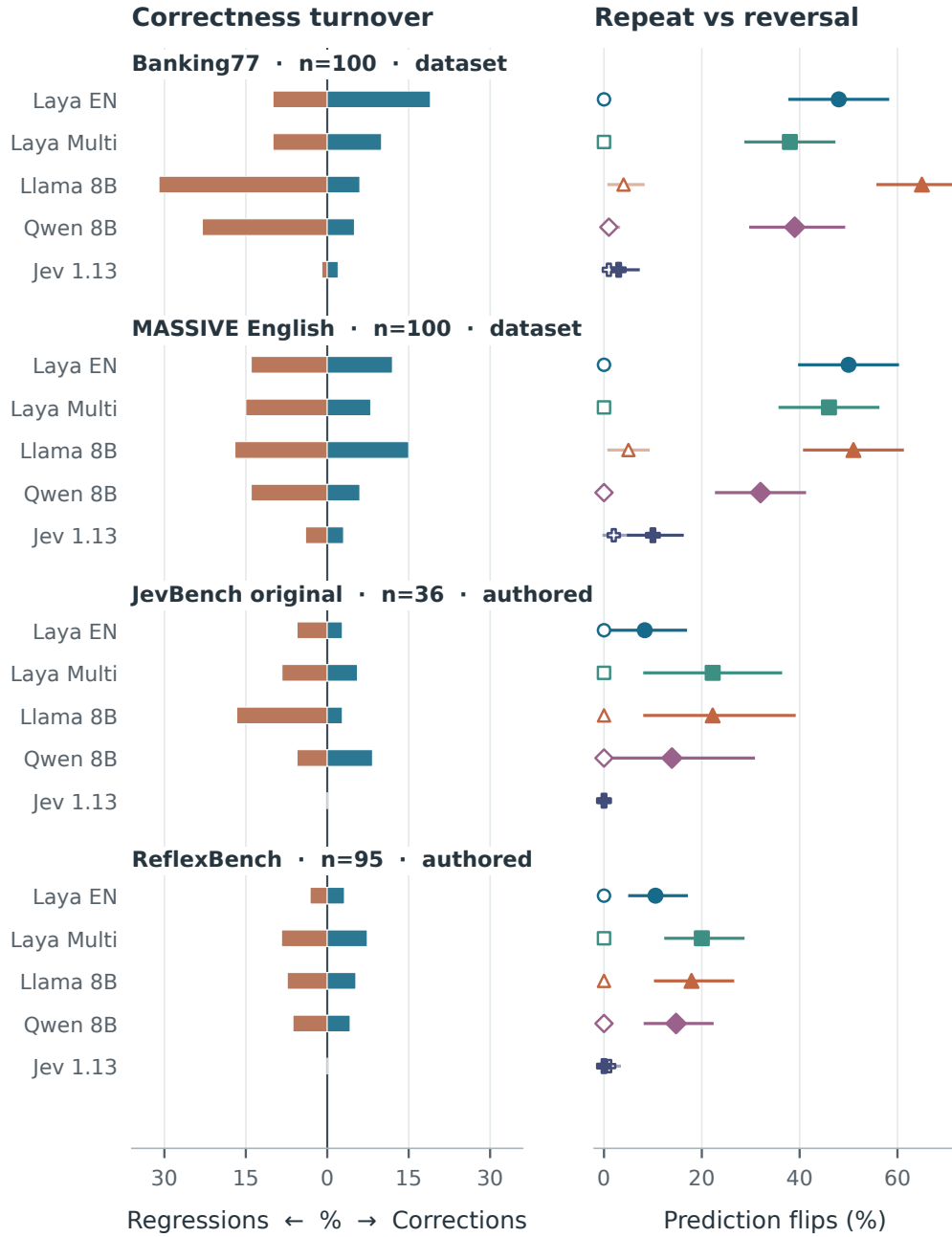


Figure 5: Historical prediction and correctness transitions under candidate reversal. Source/control denominators are 100/100, 100/100, 36/18 and 95/95 decisions/source clusters; labels have distinct provenance. Whiskers are pointwise source-cluster intervals.

B Natural-domain direct-code control

On the same 144 ContractNLI and 180 three-class SciFact cases per model, we compared the historical one-pass candidate-code logit adapter with greedy direct code generation under the unchanged historical prompt. The strict parser accepts only one code for the whole response. All 648 generated responses are valid and reproduce the corresponding semantic labels exactly: Llama has 44/144 legal and 78/180 scientific cases correct under each adapter; Qwen has 65/144 and 104/180. Within each model/domain, the paired source-cluster accuracy difference is zero with bootstrap interval $[0, 0]$, and no corrections or regressions occur. This controls a direct-prompt adapter discrepancy on these samples only. Raw responses, parsing and audit hashes are preserved in `research/natural_direct_v1/`.

C Three-class scientific codebook follow-up

This exploratory control reuses the same 180 distinct SciFact claim/document pairs as Section 4.1. The source-pair IDs and references remain fixed. Two missing display-only and code-only cells were frozen before their inference, while the already inspected base and joint cells were reused after exact tokenized-prompt hash checks. All four Qwen cells used batch size four, the pinned weights/adapters, and identical BF16/SDPA execution; all 720 responses are valid. The joint cell has no exact top-logit ties, and native and common-semantic tie decoding coincide in every cell. Correct counts for base/display/code/joint are 104/114/114/101; the interaction is $(101 - 114 - 114 + 104)/180 = -12.8$ percentage points. A 10,000-draw paired percentile bootstrap over the 180 unique claims gives $[-19.4, -6.1]$ points. This follow-up was planned after the original three-class result was visible, although the new cells were frozen before inference; it is a selected-sample interface diagnostic rather than a preregistered population interaction. Protocol, source-only freeze and all cell outputs are in `research/scifact3_codebook_v1/`.

D Full cited-pair census and run-stability sensitivity

The post-hoc census includes all 339 eligible SciFact development claim–explicitly-cited-abstract pairs: 138 SUPPORT, 71 CONTRADICT and 130 derived NOINFO. Its 159-pair complement to the earlier balanced three-choice selection contains 78/11/70 by class. This complement is not a held-out source: selected and complement pairs share 21 claims and 41 documents, and 37 complement positives appeared in our earlier two-choice task. Every one of the five systems received all four freshly generated conditions on all 339 pairs (1,356 requests/system). The freeze was fixed before this fresh inference, but the census question followed inspection of the selected-180 results. All base/reversal outputs were valid; Jev had one invalid decision under title removal only.

System	Full 339 B→R	Complement 159 B→R	Comp. NOINFO 70	Flip / repeat
Laya English	171→164	90→88	16→13	29 / 0
Laya Multilingual	150→150	77→80	8→8	63 / 0
Llama 3.1 8B	182→158	103→90	33→15	73 / 0
Qwen3 8B	231→220	129→119	60→38	97 / 0
Jev 1.13	291→292	144→143	59→59	5 / 4

Table 3: Fresh-run reference-correct counts and valid semantic base-to-reversal/repeat flips. The option reversal is a joint display-plus-answer-code change for code-likelihood adapters. The complement is within the same source, not an independent data set.

For Qwen on the complement, the paired claim-cluster 95% interval for the NOINFO accuracy difference is $[-42.4, -21.1]$ percentage points; document-cluster sensitivity gives $[-42.9, -20.3]$. The overall difference is -6.3 points with claim interval $[-13.7, +1.2]$. These are fixed-development-source, pointwise descriptive intervals, not uncertainty over scientific domains or reference validity. The 11 complement contradictions span only six claim clusters; their apparent $4 \rightarrow 7$ improvement is especially imprecise. NOINFO remains an explicitly cited pair with no annotated abstract-evidence entry, not an independently adjudicated neutral relation.

334 An overlap audit matched all 720 earlier selected-180 request hashes to the fresh census and ev-
335 ery available tokenized-prompt hash for both local LLMs. Nevertheless, two Qwen selected-180
336 base NOINFO predictions changed to SUPPORT across runs, yielding prior 104/180 versus fresh
337 102/180 base correctness; reversal stays 101/180. The batch size stayed four, but co-batches dif-
338 fered, and the audit does not identify the cause. The fresh census is therefore never assembled
339 by combining prior and complement outputs. An offline common-semantic tie-break changes full-
340 census reversed correct counts only from 158 $\rightarrow$ 159 for Llama and 220 $\rightarrow$ 219 for Qwen. Raw
341 hashes, case-level transitions, cluster resampling and reference-provenance sidecars are retained in
342 `research/scifact3_census_v1/`; access and upstream-license review are pending.

343 **D.1 Execution-context stability is not interface invariance**

344 Motivated by the cross-run drift above, we froze a new follow-up before its own inference: all 339
345 census pairs in both base and joint-reversal conditions, two local LLMs, and five execution plans
346 (6,780 decisions). The plans are original contiguous batches of four, an identical repeat in the same
347 process, reversal of row order within the same batches, seeded reassignment to new batches of four,
348 and single-request execution. No case is selected by its reference, prior error or logit margin. The
349 unchanged adapter uses the pinned weights, chat template, BF16/SDPA and explicit position IDs;
350 every unpadded prompt-token and request hash matches across plans and against the prior census.
351 Only execution context changes, not the model-facing candidate interface.

352 Both models have zero semantic flips under identical-batch repeat and within-batch row reversal.
353 Repacking changes 1 base and 4 reversed-condition Qwen predictions, versus 3 and 5 for Llama.
354 Singleton execution changes 3 and 3 Qwen predictions, versus 7 and 5 for Llama, respectively; each
355 count has denominator 339 (Figure 6). Original top-two candidate-logit margins of these changed
356 decisions are at most 0.5 for Qwen and 0.375 for Llama; several are exact BF16 ties. Native tie
357 decoding is retained. The original-plan semantic labels reproduce the previous full census for both
358 models. Nonetheless, across all five plans, joint option reversal still changes 95–97 Qwen and 71–73
359 Llama predictions. The derived NOINFO reference-agreement difference stays between -34.6 and
360 -33.1 percentage points for Qwen and between -27.7 and -26.9 for Llama. Each plan’s paired
361 claim-cluster 95% interval excludes zero; these are pointwise descriptive intervals on the fixed source,
362 not independent replications across plans.

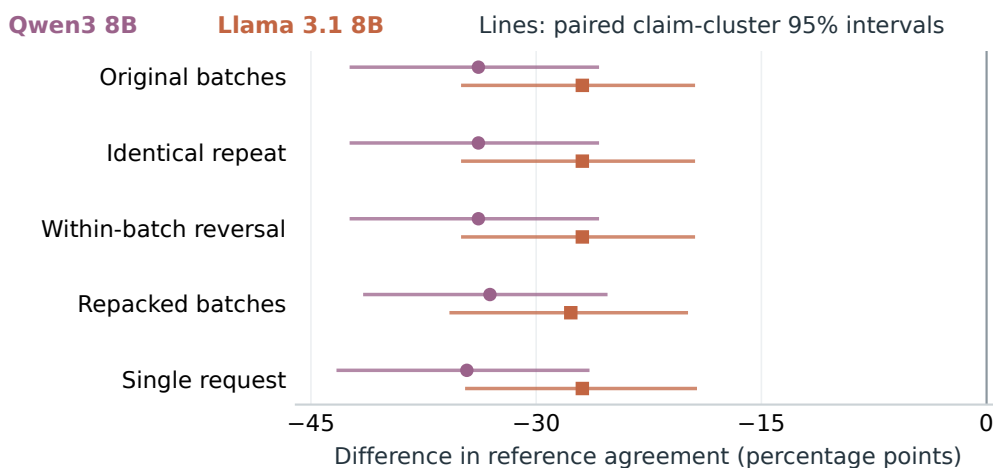
EXECUTION CONTEXT / SCIENTIFIC EVIDENCE

339 identical inputs per condition; 5 plans; 2 fixed local adapters

a Prediction flips vs. original batches (count / 339)

	Qwen3 8B		Llama 3.1 8B	
	Base	Reversal	Base	Reversal
Same batches, repeat	0	0	0	0
Same members, row reversal	0	0	0	0
New co-batches, size 4	1	4	3	5
Single-request execution	3	3	7	5

b NOINFO reference agreement: reversal minus base (pp)



NOINFO: 130 derived cited-annotation-absence references; not adjudicated neutrality.

Fixed plan order; same-source follow-up; no isolated execution-mechanism claim.

Figure 6: Execution-context follow-up on all 339 SciFact cited pairs. (a) Same-condition semantic flips relative to original contiguous batches; columns distinguish base and joint option reversal. (b) Reversal-minus-base reference agreement on all 130 derived NOINFO pairs, with 10,000-draw paired claim-cluster percentile intervals. Execution perturbations move few predictions while the class-specific interface effect persists. NOINFO is cited annotation absence, not adjudicated neutrality. Fixed execution-plan order is not counterbalanced.

363 This control supports robustness of the class-level interface pattern to the tested execution contexts,
 364 not bitwise determinism or a unique numerical mechanism. Repacking and singleton execution
 365 also change padding and kernel shapes; the fixed plan order does not isolate time-dependent effects.
 366 Same-process repeats do not establish cross-process, cross-GPU or cross-version reproducibility.
 367 These new outputs remain separate from historical results in `research/batch_context_v1/`.

368 **Reference audit prepared, not completed.** A source-verified blinded work package includes all
 369 130 derived NOINFO pairs plus 15 annotated support and 15 contradiction calibration pairs. Two
 370 independently randomized packets omit reference labels, strata and model outputs; independent
 371 annotation forms are blank. The workflow retains four-way agreement (including UNCERTAIN),
 372 separate derived/calibration statistics, locked form hashes and third-reviewer adjudication. Source-
 373 containing packets remain restricted and precisely ignored by version control. No human-validity
 374 result is claimed until independent reviewers complete this work.

E Cross-domain error complementarity

Historical cross-domain scores can conceal two different properties: whether the systems make *complementary* errors, and whether their *agreement* is trustworthy. We examine all 28 main suites without pooling across their 15 source collections or reference standards. For a suite s , let c_{im} denote reference agreement of system m on typed decision i . We define the best observed single-system rate $b_s = \max_m \text{mean}_i c_{im}$ and the reference-aware oracle ceiling $u_s = \text{mean}_i \max_m c_{im}$. Their difference $u_s - b_s$ is *selection headroom*, not the gain of a trained or deployable router: the oracle uses each decision’s reference to choose a system. Separately, a strict five-system majority exists only when at least three valid outputs are the same label. We report its coverage and conditional error risk, retaining failures in the fixed five-system panel. Pointwise intervals resample source/state clusters within each suite.

Figure 7 reveals why the two properties should not be conflated. On SST-5 with the choice interface (dataset-provided reference), the best observed system reaches 55.9% agreement and the oracle ceiling 87.3%, a 31.4-point gap (95% cluster interval [28.5, 34.4]); nevertheless, strict majority covers only 78.6% of decisions and errs on 42.7% of those covered (interval [39.3, 46.1]). On Aegis2 prompt annotations (human-reference track), oracle headroom is 16.0 points [14.4, 17.7], while majority covers all decisions but has 26.0% error [24.1, 28.0]—8.4 points less reference agreement than the best observed system (paired interval [−10.5, −6.4]). The opposite geometry also occurs: on the programmatic Jev–Laya needle suite, oracle headroom is only 0.3 points [0.0, 0.7] while majority risk is 21.3% [13.0, 30.1] at 99.8% coverage; that suite has 900 decisions but only 25 original content clusters. A high oracle ceiling thus identifies *potential* error complementarity, not a consensus-based way to realize it. Because system selection, examples and analytical questions were developed after the historical outputs existed, these are exploratory within-suite descriptions, not evidence that a cross-domain selector would generalize.

F Generative adapters expose model–interface interaction

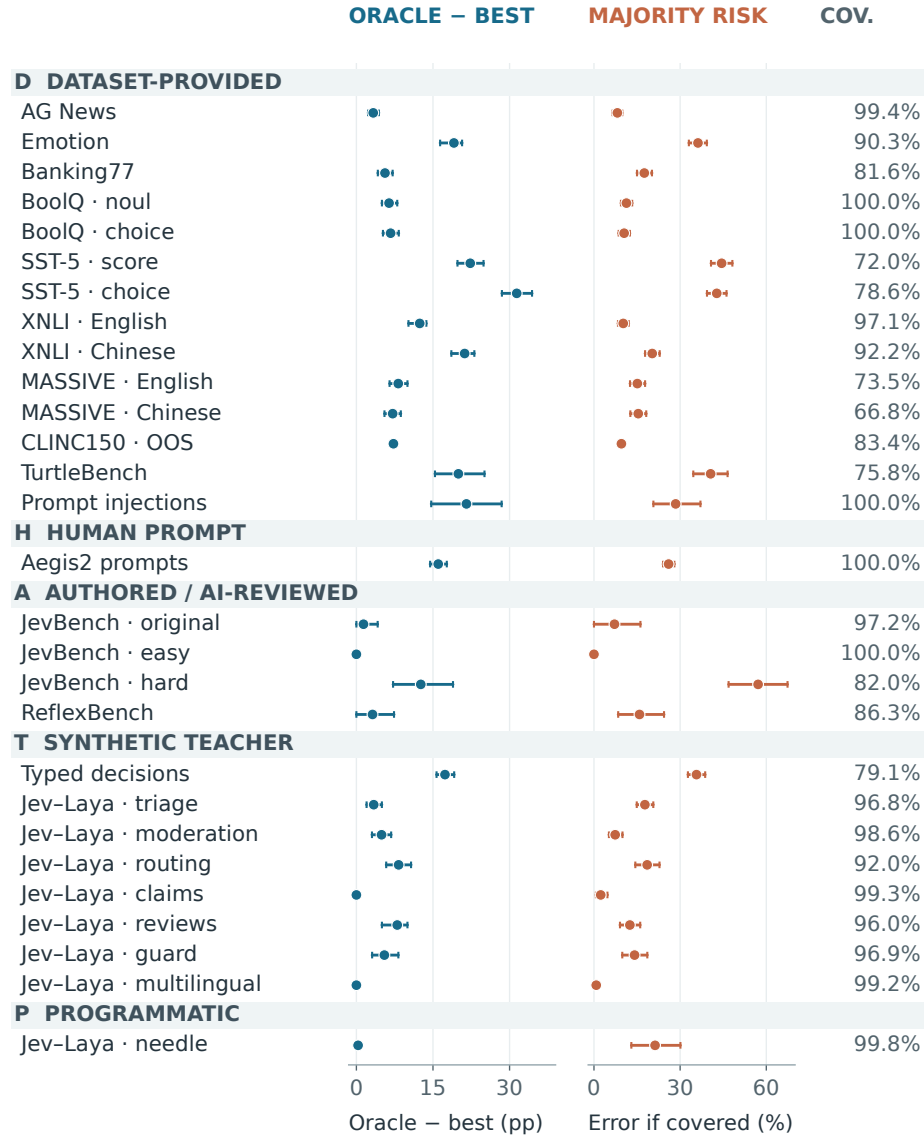
The historical Llama and Qwen comparisons use a fixed one-forward-pass candidate-code logit adapter, not free generation. To test whether that choice drives the conclusions, we froze a prediction-independent, source-balanced challenge of 212 states from 18 suites spanning all 15 current source collections. One question per state, its reference, candidate order and full input are held fixed. CLINC150/OOS is deliberately balanced between six rejection and six named-intent cases; this is a diagnostic subset, not a prevalence-weighted benchmark estimate. Each model is evaluated with two additional zero-shot adapters: greedy direct code generation using the historical prompt, and greedy generation after a fixed brief-explanation instruction ending in a final code. The checkpoint and chat-template hashes match the original baseline. Neither prompted adapter emits calibrated typed confidence or an ordinal expectation, so this comparison concerns discrete label accuracy and decision turnover only.

We also froze a smaller, exact-input direct-generation control on the new natural-domain cases: the same 144 legal and 180 three-class scientific requests for each 8B model, with the historical message template, greedy decoding and a strict whole-response one-code parser. All 648 generated answers are valid and have exactly the same semantic label as the corresponding one-forward-pass code-logit outputs; no source-domain accuracy or prediction transition changes under this adapter substitution. Prompt-token hashes and source IDs match the original requests. This rules out a “single-token scoring versus direct code generation” discrepancy on these selected cases and this prompt; it does not show that either model’s best prompted legal/scientific performance has been reached, or that their probability outputs are equivalent.

Figure 8 separates two effects that an aggregate score would hide. Direct generation changes none of Qwen’s 212 predictions and only five of Llama’s, consistent with the historical adapter’s constrained next-token decision on this subset. Brief deliberation changes 55 Llama and 46 Qwen predictions. Llama’s correctness rises from 133/212 to 145/212: 25 incorrect decisions are corrected while 13 correct decisions regress, with three invalid generations. Qwen moves from 158/212 to 153/212: 12 corrections coexist with 17 regressions and ten invalid generations. The selected-state paired effects are +5.7 points (95% interval [+0.5, +10.8]) and −2.4 points (−7.1, +2.4), respectively. However, when the 15 sources receive equal weight, Llama’s effect is +4.7 points with interval [−1.4, +10.6]; a source-general improvement is not established.

Complementary errors, fragile consensus

28 suites · 5 systems · pointwise 95% state-cluster intervals



Oracle uses each reference: upper bound, not a deployable selector.

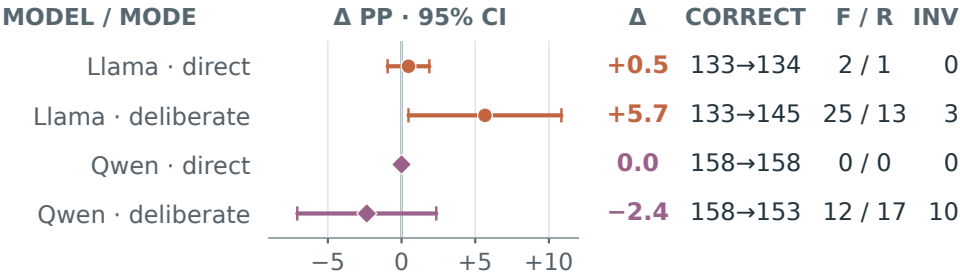
Majority: ≥ 3 matching valid labels; risk is conditional on coverage.

Figure 7: Selection headroom and strict-majority risk across 28 main suites, grouped by reference provenance: D, dataset-provided; H, human prompt; A, authored or AI-reviewed; T, synthetic teacher; P, programmatic. Left: the reference-aware oracle ceiling minus the best observed system, *not* an attainable router gain. Right: error conditional on at least three identical valid labels, with coverage in the adjacent column. Whiskers are pointwise 95% state-cluster bootstrap intervals (5,000 draws). Shared-source variants are not independent; no across-source inference is made.

Prompting changes who is right

Strict one-code parsing · matched cases · no prompt search

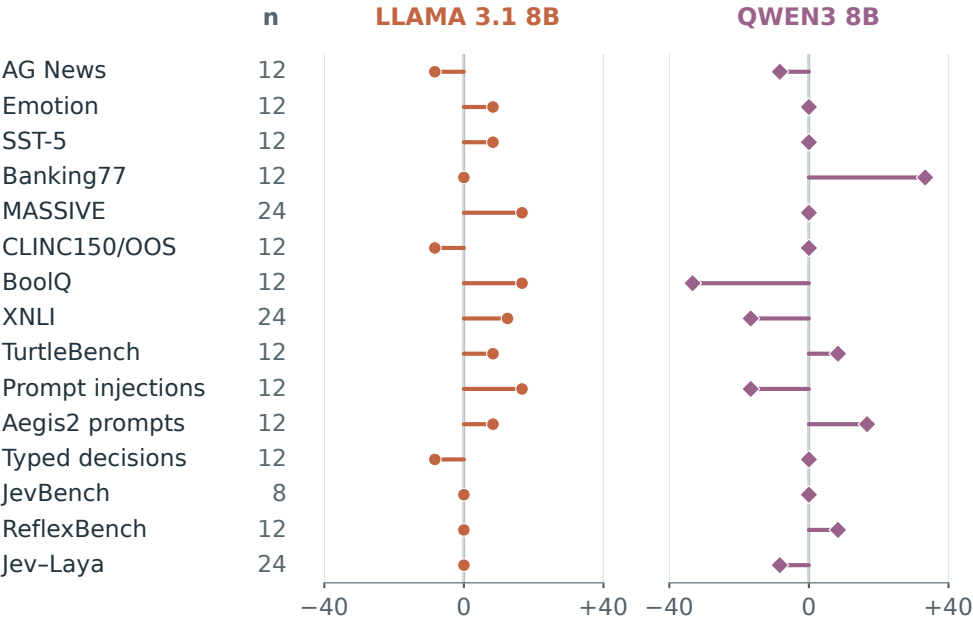
A PAIRED OUTCOMES



Suite-stratified paired CI; F/R = corrected/regressed; invalids wrong.

B SOURCE HETEROGENEITY

DELIBERATION – CODE-LOGIT · ACCURACY POINTS



Shared ±40 pp scale; source rows are descriptive (n=8-24).

Strict parser; post-hoc lenient Qwen parsing is excluded.

Figure 8: Prompted-baseline stress test on 212 matched states/model. Top: correction and regression flows, net accuracy effect and 95% paired suite-stratified bootstrap interval versus the historical code-logit output. Bottom: descriptive brief-deliberation effect by source ($n = 8-24$), on a shared percentage-point scale, without per-source significance tests. Invalid generations count incorrect; the post-hoc parser sensitivity is excluded.

429 The source view makes the interaction more concrete without elevating small cells into claims
430 of significance. Llama corrects four additional English XNLI cases but loses one Chinese XNLI

case. Qwen corrects four Banking77 cases, whereas four BoolQ answers become invalid under the predeclared output grammar; its XNLI sample loses four correct answers in 24 states. These results suggest that prompting is an intervention on a *model–adapter–task* combination, not an intrinsic improvement operator for an LLM.

Output validity materially affects the conclusion. Llama’s three invalids hit the 128-token cap before a final code. Eight of Qwen’s ten invalids actually end in an inline FINAL: code rather than the required standalone final line; two say FINAL: None. A clearly post-hoc end-of-response parser would recover eight answers, six correct, yielding 159/212 for Qwen instead of the registered 153/212. We retain the strict result as primary: the sensitivity distinguishes semantic selection from interface compliance, not permission to tune the evaluator after seeing answers. Brief deliberation used 87.6/90.1 seconds of synchronized local generation for Llama/Qwen versus 15.7/16.9 seconds for direct generation (batch size four, excluding load and tokenization); these implementation times are not merged with the separate performance experiment. One frozen 128-token prompt cannot bound either model’s best prompted performance.

G Operational measurements and confidence ranking

G.1 Local quality and completion time

The local performance experiment evaluates 11 strata, four checkpoints, three batch sizes and eight rounds on the same A100 80GB PCIe, yielding 1,056 cells and 98,304 measured decisions. Each stratum uses 64 fixed measured states and disjoint warm-up states. Model processes are fresh per block; model order is balanced, CPU affinity and thread counts are fixed, and an external observer records GPU, worker-core and SMT-sibling activity.

The timing boundary includes uncached serialization, tokenization, transfer, inference and native structured-answer construction, with GPU synchronization. It excludes model loading, disk output, validation, network transport and queueing. Batch-one p50/p95 characterize request completion; batch-eight and batch-32 work divided by prediction time characterize fixed-batch throughput. Round statistics are summarized by their median and an eight-round bootstrap interval. Repeated states increase timing replication, not the effective accuracy sample size.

Figure 13 compares quality and latency within six declared workloads. For BoolQ, median-of-round request p50 values are 10.30, 8.55, 36.86 and 33.51 milliseconds for Laya English, Laya Multilingual, Llama and Qwen respectively. The quality coordinate belongs to the same fixed workload. It is not appropriate to infer a cross-task winner from the horizontal position alone: systems can trade reference agreement for lower completion time, and that trade-off changes with the task.

The host is shared. No foreign GPU compute process or sustained CPU/SMT threshold violation was observed, but six isolated residual-CPU intervals exceeded 15% without meeting the fixed two-interval rejection rule. These intervals remain included. All raw timing records, monitor samples and code/environment hashes are retained, and all 1,056 cells independently reconcile with the generated tables.

G.2 Selective prediction within a source

Confidence-based deferral and coverage–quality analysis are established evaluation tools, including for decision models [Geifman and El-Yaniv, 2017, Ibrahim and Zaki, 2026, Hu et al., 2026]. Figure 14 shows empirical accepted-set error as coverage grows under maximum class probability in our fixed local interfaces. All tied probabilities enter together, avoiding artificial improvements from an arbitrary within-tie ordering. The curve is computed separately for each model and source; intervals at displayed landmarks condition on the selected threshold.

These curves distinguish two failure modes that accuracy alone conflates: frequent errors at all confidence levels and errors concentrated among uncertain cases. They support a workload-specific investigation of escalation policies. A threshold selected from these test curves would be a post-hoc decision rule, so a deployable policy requires a separate calibration/development split and a held-out check. Candidate-code probabilities are evaluated as observed ranking statistics, not assumed calibrated equivalents of native decision confidence.

H Separating candidate order from answer-code assignment

The legal reversal in Section 4 is a useful stress test, but it is not a single-factor manipulation for the two code-likelihood LLM adapters: reversing the serialized candidates also changes which semantic label is assigned to each answer token. We therefore completed a factorial follow-up on the *same* 144 ContractNLI cases from 91 source contracts. The original and joint-reversal outputs are reused by hash; two missing cells reverse only display order or only label-to-code assignment. The state, option meanings, gold label, model weights and adapter are fixed. Before inference, the four rendered messages and code maps were frozen and the reused cells were checked against their historical tokenized-prompt hashes. A transparent corrective amendment aligned the new inference batch size to the historical four after an initial batch-eight pilot had already been inspected; both pilots and all raw outputs remain available. This is a follow-up diagnostic of these two pinned model–adapter pairs, not a wholly prospective four-cell test or an intervention on legal reasoning itself.

Order and answer-code mapping have different effects

144 matched ContractNLI pairs/model · fixed semantic tie-break · batch 4

Llama 3.1 8B

Qwen3 8B

	BASE MAP	SWAP MAP		BASE MAP	SWAP MAP
Orig.	30.6% BASE	34.7% 125 flips	Orig.	45.1% BASE	43.8% 10 flips
Rev.	33.3% 6 flips	39.6% 54 flips	Rev.	54.2% 36 flips	58.3% 69 flips

Cells show accuracy; flips count changed semantic labels vs each model's base.

Figure 9: Orthogonal ContractNLI candidate-rendering diagnostic for Llama 3.1 8B and Qwen3 8B ($n = 144$ matched document–hypothesis cases/model). Columns change semantic label-to-code assignment; rows change serialized display order. Cell accuracy and prediction switches from base use a *common semantic tie-break* for the 2×2 comparison; this is a labeled post-hoc sensitivity, not a replacement for the historical native outputs. All cells are evaluated at historical batch size four; intervals in the text resample the 91 contracts. The source sample is deliberately class-balanced.

Figure 9 shows the common-tie comparison. The native recorded accuracy grid (base, display-only, code-only, both) is 44, 48, 50, 59 of 144 for Llama and 65, 78, 63, 83 for Qwen; all 1,152 cell decisions are valid. Yet the single-factor *prediction* transitions have nearly opposite geometry. Changing only the answer-code mapping flips 125 Llama labels (86.8%; contract-cluster 95% interval [81.1, 92.1]) but only 10 Qwen labels (6.9%; [2.8, 11.6]); changing only display order flips 6 Llama labels (4.2%; [1.4, 7.6]) but 36 Qwen labels (25.0%; [18.1, 32.2]). The selected code itself reveals why these are not equivalent perturbations: Llama selects token A in 138/144 base and 131/144 code-only cases despite the remapping, whereas Qwen shifts from token C in 122/144 base cases to token A in 126/144 code-only cases while mostly retaining its *semantic* answers. This is evidence of distinct interface coupling in the two implementations, not a general property of either model family.

There is a third interface degree of freedom: the historical joint-reversal question also reverses the decoder's first-maximum tie priority. Exact BF16 top-logit ties occur in Llama's base/display/code/both cells 4/0/9/12 times and Qwen's 0/4/0/5 times. Re-decoding stored probabilities with one fixed semantic priority (entailment, contradiction, not-mentioned) changes only the historical joint cell: Llama 59 $\rightarrow$ 57 correct and 66 $\rightarrow$ 54 base-to-joint flips; Qwen 83 $\rightarrow$ 84 correct and 64 $\rightarrow$ 69 flips. We retain the native counts above for reproducibility, but interpret the factorial grid only under this common, explicitly post-hoc tie rule. Under that convention, the joint accuracy effects are +9.0 percentage points for Llama (paired contract-cluster interval [+2.1, +16.1]) and +13.2 for Qwen ([+4.4, +21.4]). Their difference-in-differences interactions are +2.1 ([−7.0, +10.8]) and

+5.6 points ($[-2.1, +13.0]$), respectively; these intervals do not resolve an interaction. The original exact-repeat output has zero prediction changes in either model. Conversely, the batch-eight pilot changes three Qwen display-only predictions versus the matched-batch result, reinforcing why batch shape cannot be quietly varied. All intervals describe this selected, balanced, length-capped legal sample, not ContractNLI’s natural test distribution or system-level legal competence.

I Complete source-level results

Source	Laya EN	Laya Multi	Llama	Qwen	Jev
ag_news	94.3	92.4	88.9	86.8	88.6
emotion	59.1	53.0	50.5	54.8	59.0
banking77	55.3	51.2	54.1	66.8	79.9
massive_en	54.1	42.3	57.2	66.3	77.5
massive_zh	30.4	33.2	53.5	62.6	76.0
boolq	84.6	77.7	64.8	83.5	92.6
boolq_choice	83.6	77.4	68.7	83.3	92.2
xnli_en	86.0	81.7	46.2	78.7	85.9
xnli_zh	61.5	74.3	40.5	69.1	73.6
turtlenbench	42.6	41.6	42.2	44.2	74.0
sst5	34.6	29.5	32.0	45.7	56.5
sst5_choice	49.6	35.6	42.2	43.9	55.9
prompt_injections	70.7	57.8	62.9	63.8	76.7
aegis2_prompt	49.6	57.1	66.8	72.7	82.4
typed_decisions	36.4	34.9	51.2	55.5	73.9
JB: original	70.8	41.7	75.0	83.3	98.6
JB: easy	95.8	89.6	100.0	100.0	100.0
JB: hard	29.7	32.4	34.2	46.8	72.1
ReflexBench	58.9	46.3	60.0	84.2	93.7
JL: triage	62.5	55.7	75.6	80.0	90.0
JL: moderation	67.8	48.4	89.0	77.5	92.7
JL: routing	64.2	51.1	61.3	83.7	88.8
JL: claims	90.0	80.0	61.7	98.7	100.0
JL: reviews	70.7	42.3	83.0	87.0	87.7
JL: guard	60.6	33.2	88.0	83.2	93.8
JL: multilingual	57.8	62.5	94.5	98.4	100.0
clinc150_oos	55.7	64.8	55.2	67.8	89.5
JL: needle	50.2	47.4	77.9	87.8	95.2

Table 4: All 28 main suites, in fixed taxonomy order. Entries are percentage reference agreement; synthetic/authored sources retain their respective reference semantics. These rows are not independent benchmarks and are not averaged. Failure cases remain in denominators. Full per-source cluster intervals, type/language slices and provenance are provided by the machine-readable artifact.

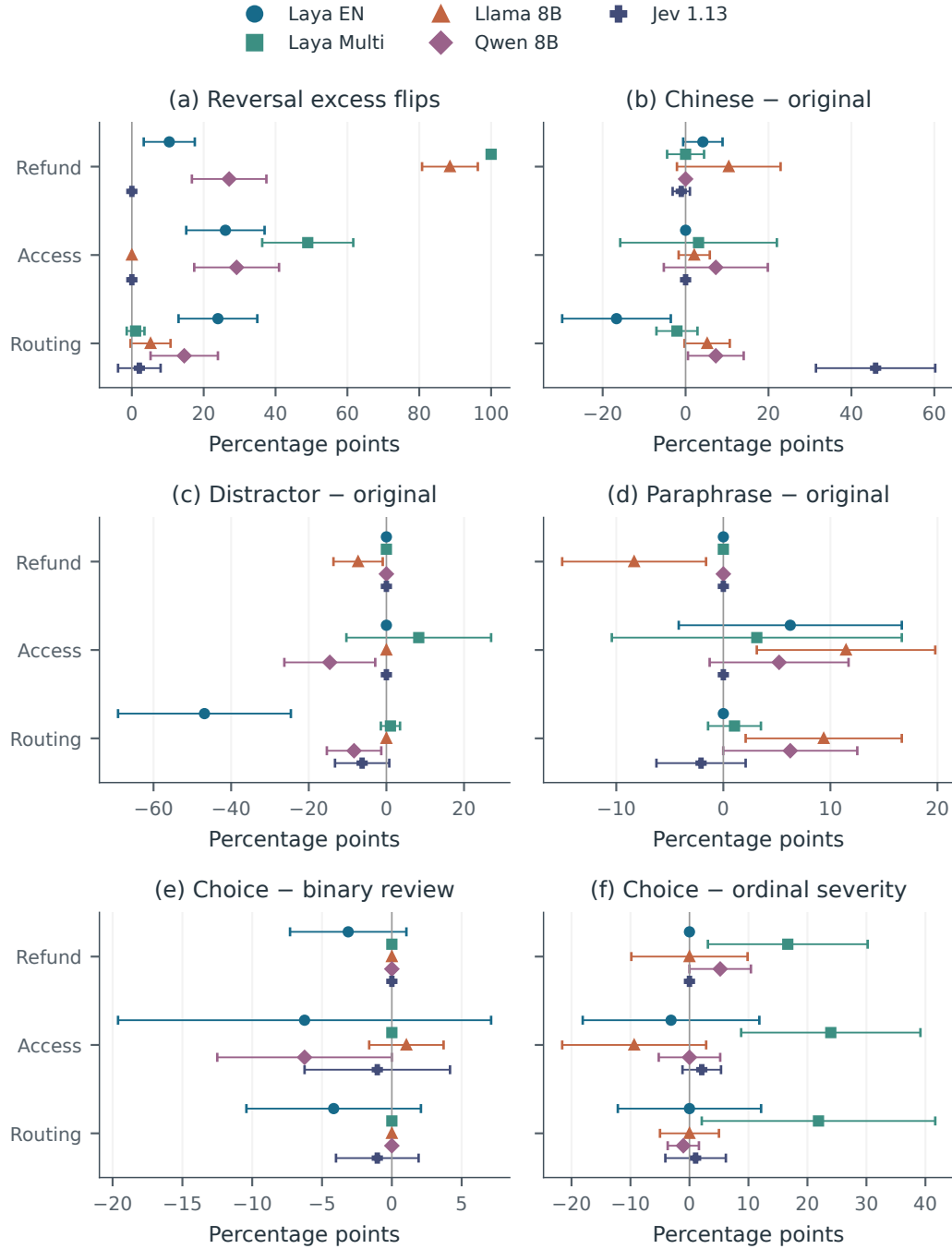


Figure 10: All six predeclared fresh-policy effects, with 96 base clusters per policy. Whiskers are simultaneous 95% intervals within the six-effect family of each model/policy, using 10,000 bootstrap draws stratified by generator seed. They do not provide paper-wide coverage. Panel (a) measures excess action flips; (b)–(f) measure correctness differences for the stated primitive. Zero-variance intervals are descriptive, not proof of invariance. Programmatic references follow visible policies. Chinese policy/question/criteria text is authored; structural field and label IDs remain in English.

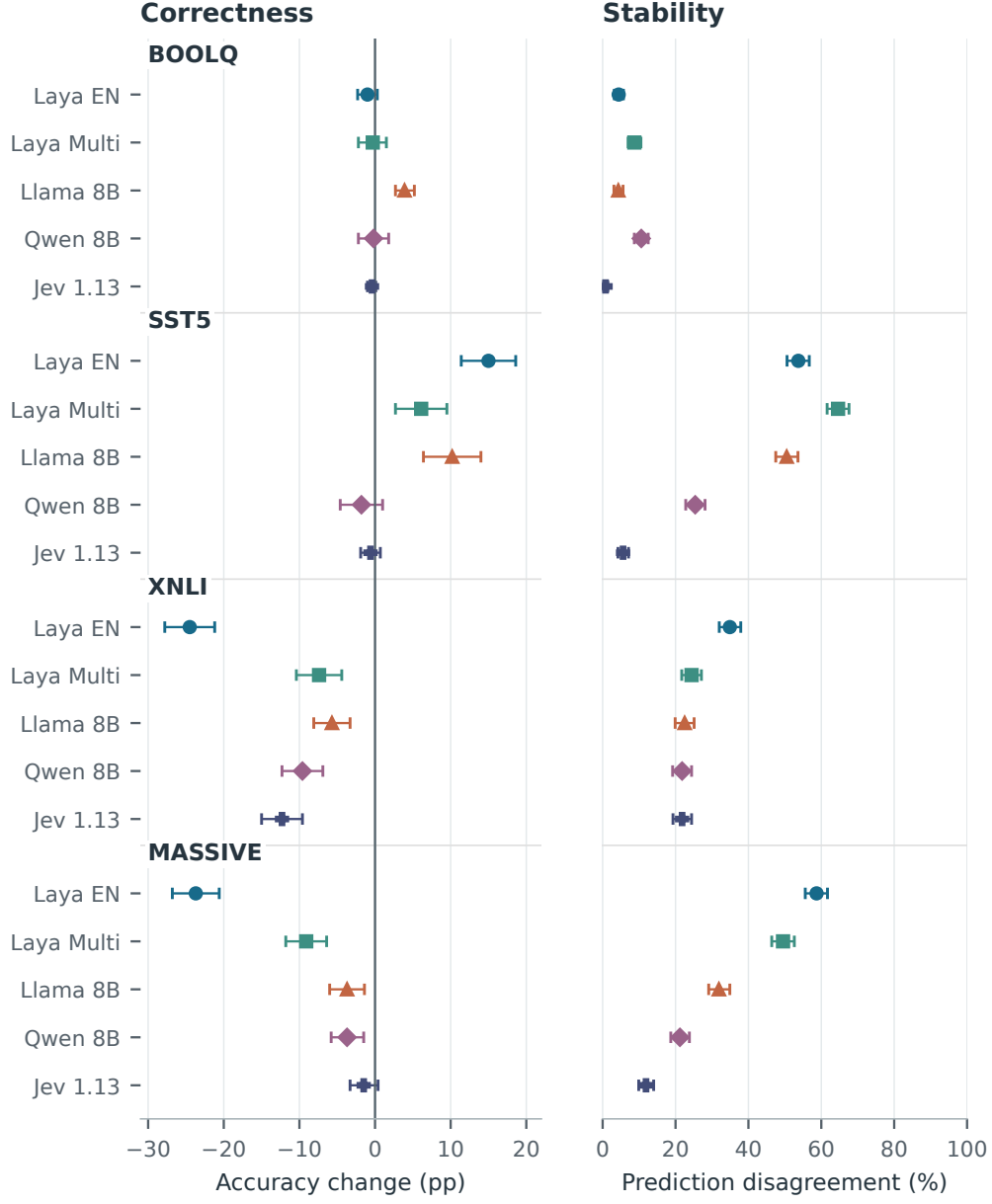


Figure 11: Exploratory paired interface contrasts against dataset-provided references on 1,000 aligned states per task: BoolQ choice minus noul, SST5 choice minus score, and Chinese minus English for XNLI and MASSIVE. Left: accuracy change; right: canonicalized prediction disagreement. Whiskers are pointwise 95% paired state-cluster bootstrap intervals (10,000 draws). Canonical label mappings are recorded explicitly; no marginal-CI subtraction is used.

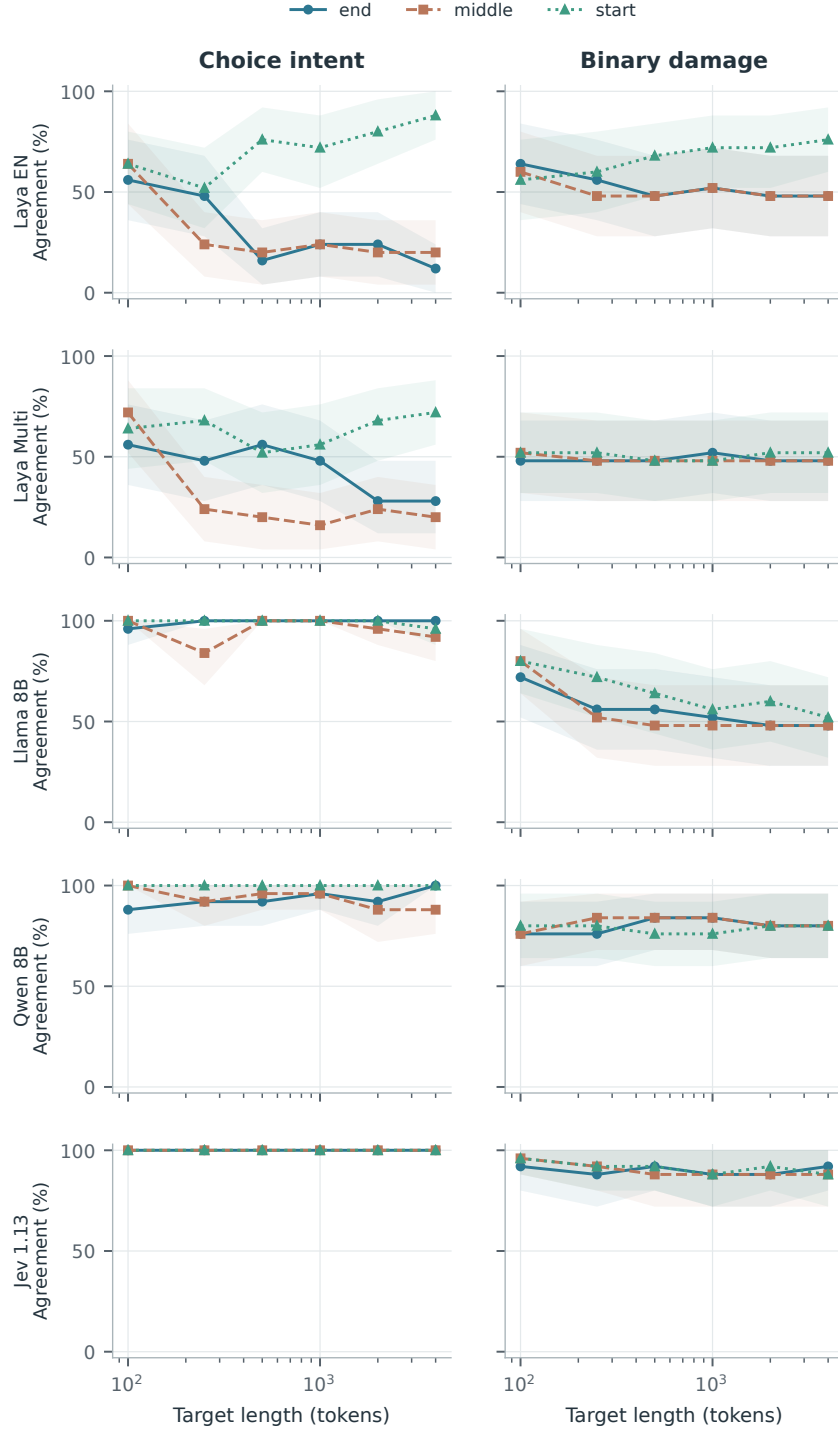


Figure 12: Programmatic needle reference agreement by source target length, insertion position and question. Rows identify models; left/right columns are the choice intent and binary damage questions. Each point uses 25 original needle clusters; shaded pointwise 95% cluster intervals resample those originals. Source length targets are not identical tokenizer token counts. The two primitives need not share the same context trend.

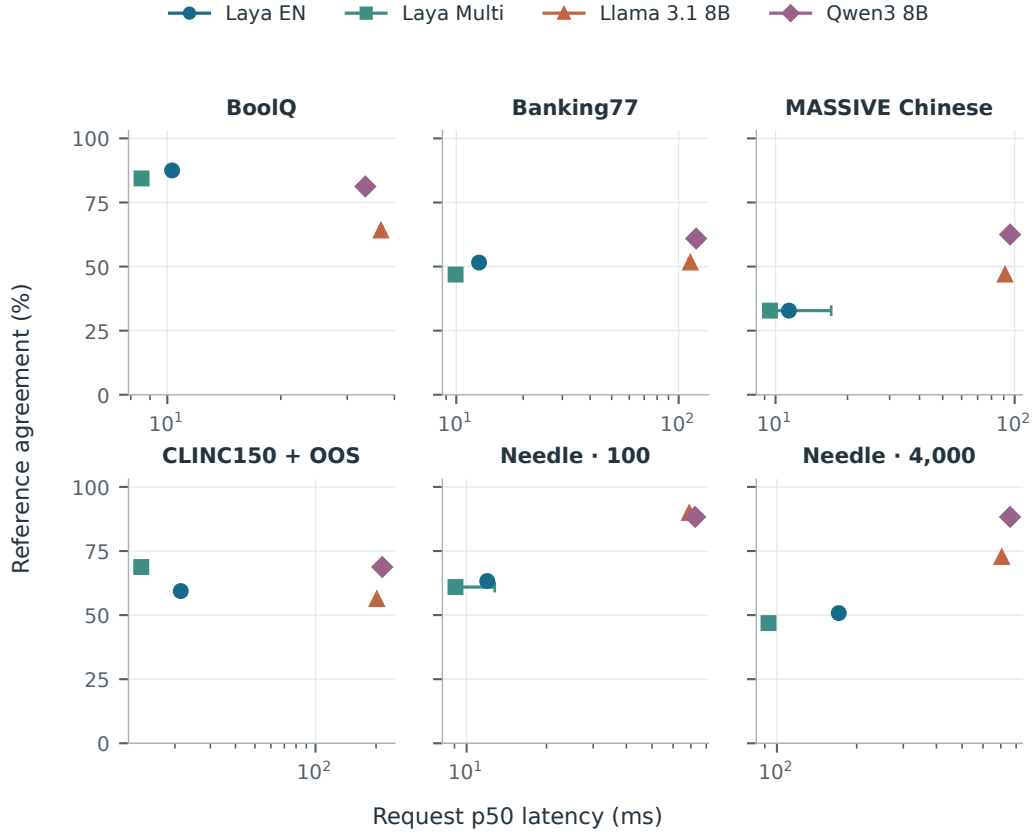


Figure 13: Resident-GPU quality/latency observations on six named workloads. Each stratum has the same 64 measured states across local models. BoolQ, Banking77, MASSIVE and CLINC150 use dataset references; the two needle workloads use programmatic references. Horizontal bars are 95% bootstrap intervals for the median of eight round p50 latencies; vertical coordinates are median reference agreement over repeated fixed states, not a new population-accuracy estimate. Jev’s hosted latency is excluded because its measurement boundary differs. No cross-workload frontier is drawn.

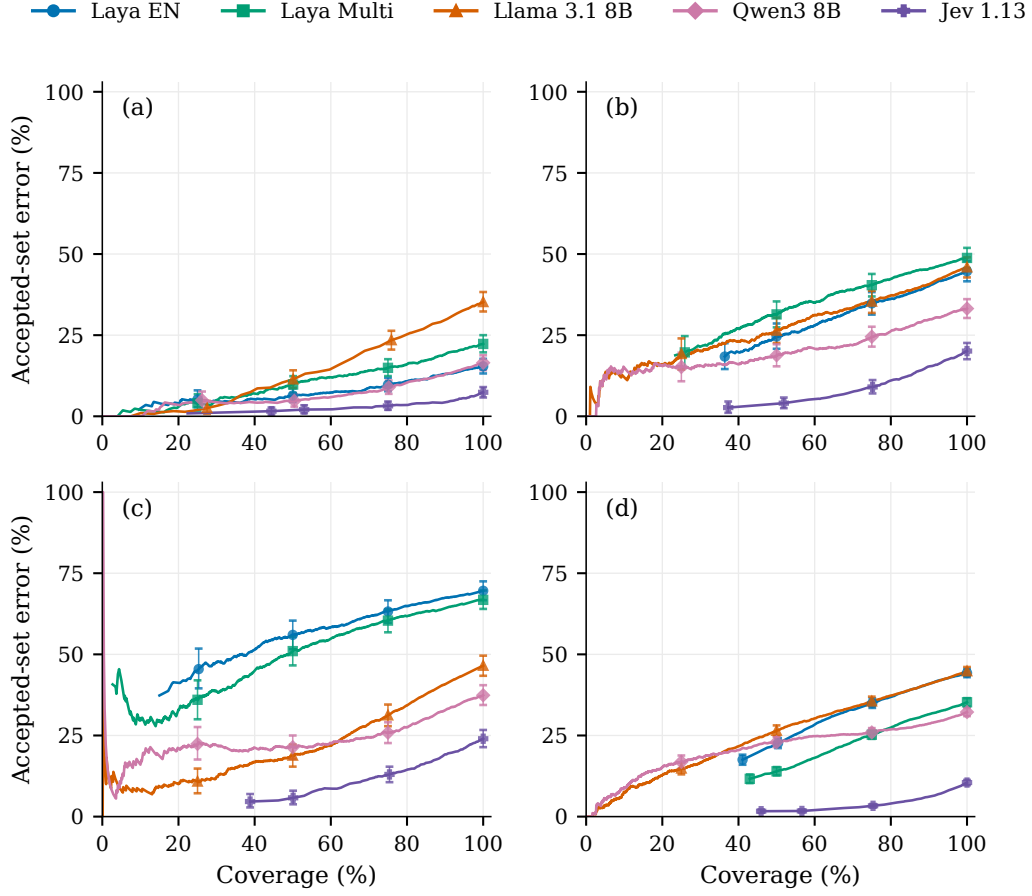


Figure 14: Maximum-probability selective error on (a) BoolQ, (b) Banking77, (c) MASSIVE Chinese and (d) CLINC150 with OOS (1,000, 1,000, 1,000 and 5,500 states). Ties are accepted together. Markers identify realized coverage at nominal 25/50/75/100% landmarks; intervals resample accepted state clusters conditional on the displayed threshold. Native and candidate-code probability semantics remain distinct. Curves describe the observed test set and do not validate a deployment threshold.

J Natural-domain sampling and reference validity

The separate legal and scientific diagnostic is deliberately balanced for paired sensitivity analysis, not a claim of natural prevalence. ContractNLI has 123 test contracts and 2,091 annotated document-hypothesis decisions, of which 220 are contradictions. The 16,000-character cap removes 24 contracts, disproportionately SEC-HTML files (13/23), versus 7/76 PDFs and 4/24 SEC-text files. The frozen 144 pairs span 91 contracts and contain 48 examples of each of the three labels; 43 of its 48 contradiction cases use four of the 17 fixed hypothesis IDs. To quantify that prior without test-label training, we compute the modal label for each hypothesis on the official 423-contract training split and apply the resulting ID-only lookup to the frozen sample: $82/144 = 56.9\%$ reference agreement (descriptive 95% contract-cluster interval [49.3, 64.3]). The same lookup reaches $1,379/2,091 = 65.9\%$ on the full official test set, illustrating how the frozen class balance changes the question. This supervised label prior is not a fair zero-shot model baseline and does not prove any system uses the shortcut. Neither the cap nor the balanced label mixture supports a prevalence-weighted comparison with ContractNLI’s published full-test scores. The observed correctness and reversal effects apply to complete selected contracts and their specified hypotheses only.

SciFact’s cited-positive subset contains 120 claim–abstract pairs from 120 distinct claims but only 111 distinct abstracts. It samples 60/64 eligible contradiction claims and 60/124 eligible support claims, so labels are balanced but claim creation mechanisms and source-paper repetition are not. The 112 development-set claims without positive evidence are excluded; the experiment neither assigns them a no-information target nor measures retrieval. Claim-cluster intervals in Figure 1 account for claim identity, not shared-abstract dependence. Re-clustering on the 111 abstracts gives nearly identical descriptive intervals: Qwen’s reversal accuracy change is -6.7 points with claim-cluster interval $[-12.5, -0.8]$ and abstract-cluster interval $[-12.7, -0.8]$; all five alternatives are recorded in the companion analysis. The supplied title and full abstract are visible, while evidence sentence indices and references are not. No local full-input audit can verify hosted server tokenization.

The first hosted attempt at this frozen experiment stopped after 216 requests when one HTTP-200 response selected a different option from its own maximum-probability option. Its append-only journal is retained as a separate aborted run. The primary Jev record is a fresh 792-request execution with an explicitly versioned runner that records such invalid typed outputs without silently selecting an alternative; no invalid decision occurred in that complete execution. This deviation is not used to replace a failed pair in the primary run. Source-text redistribution is separately governed by ContractNLI and SciFact/S2ORC notices in `research/domain_expansion_v1/DATA_LICENSES.md`; the local source-containing freeze is not shipped in Git.

Three-class SciFact source and input audit. We reconstructed the labeled development claims and complete abstracts from the official SciFact archive, binding archive, source-file, preparer, frozen-request and raw-output SHA-256 hashes in the companion artifact. Among 300 claims and 5,183 corpus abstracts, 340 citation-list occurrences reduce to 339 unique claim–document pairs because claim 1245 repeats one cited ID. The resulting pair counts are 138 SUPPORT, 71 CONTRADICT and 130 derived NOINFO; the latter require explicit membership in `cited_doc_ids` and absence of an evidence entry. Every cited ID exists in the corpus, every evidence ID belongs to its claim’s citation list, and we found no empty evidence lists, contradictory rationale labels, empty selected texts, or out-of-range rationale indices. Thirteen claims have evidence for one cited document and no evidence for another; these are different pairs, not conflicting labels. The selected 60/60/60 pairs have both unique claim and document IDs, so the 10,000-draw paired, label-stratified cluster bootstrap resamples the same 180 units across conditions. A separate source validator recomputes every selected reference and 720 request hashes; an independent analysis checker recomputes all raw counts and confusion cells. All four local adapters consumed all 720 requests without tokenizer truncation, and no system returned an invalid choice. The hosted server’s own tokenization remains unverified. SciFact claims/annotations are CC BY 4.0 and the S2ORC abstract corpus is ODC-By 1.0 under the upstream license notice; reconstructed upstream text is Git-ignored rather than redistributed.

J.1 SciFact3 codebook audit and decoder sensitivity

The exploratory four-cell diagnostic in Section 5 uses exactly the 180 previously selected, class-balanced SciFact3 claim–cited-abstract pairs. There is one pair per source claim and cited document. The observed base and joint-reversal records were reused by immutable file hashes. Display-only and

code-only requests were frozen before those missing cells were run; there was no SciFact3 factorial pilot or amendment. Before new inference, every reused prompt matched its original token sequence and length under the pinned tokenizer; checkpoint-file, chat-template, adapter, case/request, and batch-size-four hashes matched. The only model-facing user payload fields are `state`, `question_type`, `instructions`, and `candidates`. The state necessarily contains claim, paper title, and cited abstract, but not reference labels, source split, group, or source identifiers.

Model	Decoder	Base	Display only	Code only	Both
Llama 3.1 8B	native	78 (—)	61 (51)	71 (85)	69 (42)
	common tie	78 (—)	61 (51)	71 (85)	68 (46)
Qwen3 8B	native	104 (—)	114 (30)	114 (18)	101 (60)
	common tie	104 (—)	114 (30)	114 (18)	101 (60)

Table 5: Correct predictions out of 180; parentheses show semantic prediction flips relative to base. Native retains the original historical decoder, whereas common tie applies the fixed SUPPORT-CONTRADICT-NOINFO semantic priority to all four cells. Invalid counts are zero in every cell.

Exact maximum-logit ties in base/display/code/both number 6/1/11/6 for Llama and 1/6/1/0 for Qwen. Relative to native decoding, common tie changes only six Llama joint-reversal decisions; the other seven model-cell combinations are unchanged. Repeating the original base execution produced zero prediction flips for either model. An independent standard-library replay verified the SHA-256 of all eight original/new gzip result files, reconstructed all four message and candidate-assignment hashes per case, and recovered every native/common accuracy count and full base-to-cell transition matrix directly from raw stored probabilities. It also checked all reference and source-group associations without emitting source text.

Qwen3 reference class	Base	Display only	Code only	Both
SUPPORT	45/60	48/60	50/60	53/60
CONTRADICT	5/60	18/60	12/60	18/60
NOINFO	54/60	48/60	52/60	30/60

Table 6: Qwen3 correct counts by operational reference label under the common tie decoder (identical here to native decoding). The NOINFO reference means no annotated evidence in the selected cited abstract, not evidence of absence in the full paper or literature.

For the descriptive accuracy interaction ($\text{both} - \text{code}$) - ($\text{display} - \text{base}$), the Qwen estimate is -12.8 percentage points with a pointwise 95% percentile interval $[-19.4, -6.1]$; Llama under common tie is $+7.8$ points $[-1.1, +16.7]$. Both intervals use 10,000 paired bootstrap draws with seed 20260930, re-sampling source claims and retaining all four matched decisions together. Because each selected claim contributes only one pair, claim-cluster and case resampling coincide here. These are exploratory post-hoc-to-the-existing-outcomes, unadjusted, selected-sample intervals; they are neither population-prevalence estimates nor neural-mechanism identification. The full protocol, frozen ID/hash manifest, original raw results and replay code are under `research/scifact3_codebook_v1/`; no upstream abstract text is duplicated in its freeze.

K Reproduction and data accounting

The historical accuracy record contains 22,934 logical requests and 26,450 decisions per model, including matched controls. The timing record contains 67,584 measured request executions and 98,304 decisions across repetitions. The fresh policy diagnostic contains 288 base states, 2,304 request variants and 6,912 decisions per evaluated model. The held-out wording test adds 96 new base states and 384 requests per model. These totals refer to different collections and replication units and should not be combined into an independent sample size.

The repository retains source/revision manifests, prepared-input fingerprints, model metadata, raw per-question probabilities and predictions, input audits, and complete timing/monitor records. The paired-analysis script checks raw-file fingerprints and reference alignment before deriving effects. Figure scripts consume its JSON output or the verified performance summary, producing vector

606 PDF/SVG and PNG previews. The hosted runner reads credentials outside the repository, records
607 every response attempt without authorization headers, pins the returned model ID, and resumes only
608 under an unchanged contract.

609 Accuracy and typed-response validation can be recomputed on CPU from the retained raw out-
610 puts, subject to an approved artifact-access and source-license plan; no public release is claimed.
611 New model inference additionally requires the pinned checkpoints or a provider account. Exact
612 remote replication depends on the provider honoring its version identifier. Generator code and full
613 programmatic confirmation cases are reproducible without downloading third-party state text.

614 **Hosted execution deviation.** The frozen retry description referred to 5xx errors generally, while the
615 implementation enumerated HTTP 500, 502, 503, 504 and 529 (plus 429 and transport exceptions).
616 A single HTTP 520 in the fresh refund Chinese condition received one attempt and caused three
617 failed decisions, retained in primary denominators. Its Chinese-minus-original action effect is
618 $-1/96$; a valid-pair sensitivity excludes that paired base state and yields $0/95$. This service failure
619 does not support a language-regression claim. The historical matrix additionally contains four
620 choice/probability-consistency failures affecting eight decisions, also retained. No failed answer was
621 repaired or replayed.

622 One historical resume boundary also violates the protocol’s global manifest-order statement: two
623 adjacent journal entries cross that ordering. The journal remains append-only, and keyed reconciliation
624 verifies complete request identities, payloads and outputs. No scoring or selection depends on journal
625 position; the original bytes and the execution deviation are preserved.

626 **L Prospective submission-readiness note**

627 This document is an internal, double-blind layout rehearsal, *not* a NeurIPS 2026 submission; that
628 cycle’s deadline has passed. A future target year’s format, page limit and official checklist must be
629 checked afresh and completed with evidence. No public benchmark release, artifact-access review, or
630 checklist compliance is claimed here. Before submission, an anonymized code/data artifact needs
631 approved scope, upstream-license compliance and an access plan; the source-containing ContractNLI
632 and SciFact freezes remain local under their respective redistribution notices. If the venue requires
633 new-dataset metadata such as Croissant, that must also be prepared. Collaborator or independent
634 human adjudication is still needed before claiming that cited-but-unannotated NOINFO pairs are truly
635 neutral or that bilingual policy renderings are semantically equivalent. Hosted-model replication
636 additionally depends on provider access and the stability of its returned version identifier.